$$x = \{1, 2, -1, 3, 1, 0, 2\}$$

$$h = \{1, 2, -1\}$$

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]\, h[n-k]$$

$$\uparrow \mathcal{H}$$

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\, \delta[n-k]$$

		1	2	−1	3	1	0	2			
−1	2	1									= 1
	−1	2	1								= 4
		−1	2	1							= 2
			−1	2	1						= −1
				−1	2	1					= 8
					−1	2	1				= −1
						−1	2	1			= 1
							−1	2	1		= 4
								−1	2	1	= −2

$$y = \{1, 4, 2, -1, 8, -1, 1, 4, -2\}$$

$$(x * y)[n] = \sum_{k=-\infty}^{\infty} x[k]\, y[n-k] =$$

$$(y * x)[n] = \sum_{k=-\infty}^{\infty} y[k]\, x[n-k]$$

$$m = n - k$$

$$k = n - m$$

$$= \sum_{m=\infty}^{-\infty} y[n-m]\, x[m] = \sum_{m=-\infty}^{\infty} x[m]\, y[n-m]$$